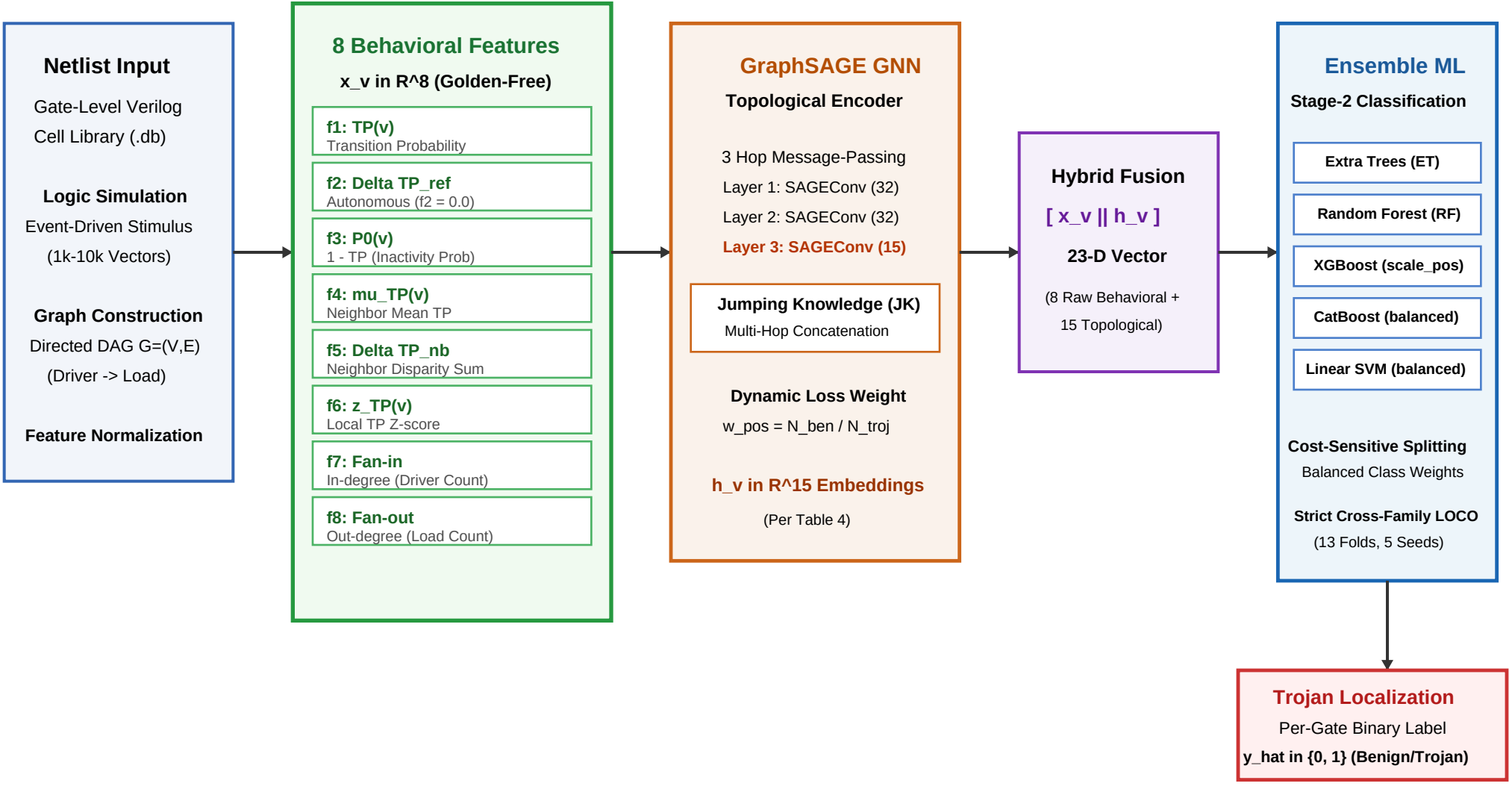
Netlist Input
Gate-Level Verilog
Cell Library (.db)
Logic Simulation
Event-Driven Stimulus
(1k-10k Vectors)
Graph Construction
Directed DAG G=(V,E)
(Driver -> Load)
Feature Normalization
8 Behavioral Features
x_v in R^8 (Golden-Free)
f1: TP(v)
Transition Probability
f2: Delta TP_ref
Autonomous (f2 = 0.0)
f3: P0(v)
1 - TP (Inactivity Prob)
f4: mu_TP(v)
Neighbor Mean TP
f5: Delta TP_nb
Neighbor Disparity Sum
f6: z_TP(v)
Local TP Z-score
f7: Fan-in
In-degree (Driver Count)
f8: Fan-out
Out-degree (Load Count)
GraphSAGE GNN
Topological Encoder
3 Hop Message-Passing
Layer 1: SAGEConv (32)
Layer 2: SAGEConv (32)
Layer 3: SAGEConv (15)
Jumping Knowledge (JK)
Multi-Hop Concatenation
Dynamic Loss Weight
w_pos = N_ben / N_troj
h_v in R^15 Embeddings
(Per Table 4)
Hybrid Fusion
[x_v || h_v]
23-D Vector
(8 Raw Behavioral +
15 Topological)
Ensemble ML
Stage-2 Classification
Extra Trees (ET)
Random Forest (RF)
XGBoost (scale_pos)
CatBoost (balanced)
Linear SVM (balanced)
Cost-Sensitive Splitting
Balanced Class Weights
Strict Cross-Family LOCO
(13 Folds, 5 Seeds)
Trojan Localization
Per-Gate Binary Label
y_hat in {0, 1} (Benign/Trojan)